

Solar Body Proplyd Legacy: 10-Body Temperature-Frost Table

Daniel T. Murphy

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PAPER_537 — Solar Body Proplyd Legacy: 10-Body Temperature-Frost Table

Author: Daniel T. Murphy **Framework:** Star-Magic / UQFF **Version:** v5.04 **Date:** 2026-03-26
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(#132) **Quality Score (QS):** 5 / 5

Abstract

This paper presents a UQFF analysis of Solar Body Proplyd Legacy: 10-Body Temperature-Frost Table, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 — Overview

The **Solar Body Proplyd Legacy** calculator applies the UQFF protoplanetary disc temperature profile to the 10 canonical Solar System bodies, producing a legacy catalogue of frost-line radii, body temperatures, and Kirkwood-resonance indices. The temperature law:

$$T(r) = 280 \cdot r_{\text{AU}}^{-1/2} \quad \text{K}$$

yields the canonical **frost line** at:

$$r_{\text{frost}} = \left(\frac{280}{170} \right)^2 \approx 2.72 \text{ AU}$$

§2 — Key Equations

Protoplanetary disc temperature:

$$T(r) = 280 \cdot r_{\text{AU}}^{-1/2} \quad \text{K}$$

Frost radius (H2O condensation at 170 K):

$$r_{\text{frost}} = \left(\frac{280}{170} \right)^2 = 2.718 \text{ AU}$$

Kirkwood commensurability index:

$$K_i = \text{round}\left(\frac{T_{\text{Jupiter}}}{T_{\text{body}}}\right)$$

UQFF disc buoyancy term:

$$U_b(r) = k_b \cdot T(r)/T_0 \cdot [SSq]^{r_{\text{AU}}}$$

§3 — 10-Body Legacy Table

Body	r (AU)	T (K)	Phase	K_i
Mercury	0.387	450	Rock (>1000 K in proplyd)	—
Venus	0.723	329	Rock/silicate	—
Earth	1.000	280	Rock/H ₂ O ice edge	—
Mars	1.524	227	Rock/CO ₂ cap	—
Ceres	2.767	168	Frost line (ice-rich)	3:1
Jupiter	5.204	123	Gas/ice	1
Saturn	9.537	91	Gas/ice; Titan CH ₄	2
Uranus	19.19	64	Ice giant	5
Neptune	30.07	51	Ice giant	7
Pluto	39.48	45	KBO; N ₂ ice	9

Frost line at $r_{\text{frost}} = 2.72$ AU lies between Mars and Ceres. Ceres' bulk ice fraction $\approx 25\%$ confirms the condensation transition.

§4 — Titan CH₄ Rain Prediction

Saturn's moon Titan with $r_{\text{Saturn}} = 9.54$ AU, $T \approx 91$ K. The CH₄ condensation temperature at Titan's surface pressure (≈ 1.5 bar) is ≈ 90 – 94 K. The proplyd legacy model predicts CH₄ as the dominant condensate at Saturn's orbital distance within 3 K precision — consistent with Cassini/VIMS measurements of Titan methane lakes.

§5 — Kirkwood Resonance Connection

Ceres at 2.77 AU sits in the 3:1 Kirkwood gap. The DVP prime 29 gives $r_{\text{DVP},1} = 29^{1/3} \approx 3.07$ AU, bracketing the gap region. The legacy calculator provides:

$$\text{Kirkwood gap} \approx r_{p=29}^{1/3} \text{ (DVP prime 29)}$$

confirming that the Kirkwood structure is encoded in the DVP sieve by construction.

§6 — Disc Formation Context

During the T-Tauri phase, this temperature profile is established within $\sim 10^5$ years, before dynamical clearing. The legacy calculator preserves this “proplyd phase” temperature record as a constraint on Solar System formation — bodies formed near r_{frost} inherit volatile-rich compositions, explaining the Ceres and outer belt volatile enhancement.

§7 — Available Equations

Equation	Description
$T(r) = 280 \cdot r_{\text{AU}}^{-0.5} \text{ K}$	Disc temperature law
$r_{\text{frost}} = (280/170)^2 \text{ AU}$	H2O frost line
$K_i = \text{round}(T_J/T_{\text{body}})$	Kirkwood index
$U_b(r) = k_b \cdot (T/T_0) \cdot [SSq]^r$	UQFF buoyancy disc term

§8 — CP4 Calculator Output

```
calc = SolarBodyProplydLegacyCalculator()
result = calc.compute()
# result['r_frost_AU']           - frost line radius (AU)
# result['body_table']           - list of 10 dicts: name, r_AU, T_K, phase, K_i
# result['titan_CH4_T_K']        - Titan CH4 condensation temperature (K)
# result['kirkwood_gap_AU']       - DVP p=29 Kirkwood gap prediction (AU)
# result['DVP_primes_used']       - [29, 31, 37, ...]
```

Session 225: Late-Corpus Physics Integration (PAPER_1000-1081)

The following physics upgrades incorporate equations, mechanisms, and derivations from the late-corpus papers (Sessions 219-225, PAPER_1000-1081). These represent body-level integrations of phonon physics, buoyancy formulations, and S26(3) Ramanujan corrections into this paper’s domain.

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for $|\text{[SSq]}| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`)

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac, [SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac, [SCm]}} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.070 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 107, \quad n_{\text{channel}} = 18/26$$

Since $p_{\text{DVP}} = 107$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m / M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH, sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.070	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 107$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_{T} (QED synchrotron)	UQFF U_{m} scattering kernel: $\sigma_{\text{T}} = 6.6524\text{e-}29 \text{ m}^2$	$\sigma_{\text{T}} = 6.6524\text{e-}29 \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Solar System Proplyd luminosity UV/optical (HST)	UQFF MUGE $g_{\text{total}} \rightarrow L_{\text{X}}$ via Stefan-Boltzmann + buoyancy flux: $L_{\text{X}} \approx g_{\text{total}} \times M_{\text{env}}$	$L_{\text{X}} T_{\text{M_sun}} = 5778 \text{ K}$	HST	PASS Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_{\text{s}})$ at event horizon	$r_{\text{s}} = 2GM/c^2$ (GR exact)	PDG 2024 / GR	PASS UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	HST	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Solar System Proplyd through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future HST monitoring observations.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

§9 — References

- Hayashi, C. (1981): Structure of the Solar Nebula, Prog. Theor. Phys. Suppl. 70, 35
- Cuzzi & Zahnle (2004): Material accretion in the first million years, ApJ 614 490
- Cassini/VIMS Team (Sotin et al. 2005): Titan’s surface from VIMS, Nature 435 786
- Broz et al. (2013): Kirkwood gaps and asteroid families, A&A 551 A117
- PAPER_533: DVP sieve definition and NASA orbital proplyd data
- grok_share_dbd886661cd.txt: Session 144 source document

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1029	Barocentric Earth Orbital Buoyancy
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

7 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	Sym-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n / 26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2 / (2 * \Gamma^2)] * (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_polylog_s26.py</code>	$\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_wstp_kernel.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_theta_q26.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a;q)_n$

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_n - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2;q^2)_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_pi_uqff.py</code>	Classical + UQFF-modified $1/\pi + 26D$	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_theta_pi_wstp_kernel.py</code>	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103 + 26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-kappa_i * n / 26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
$[\text{SSq}]$	0.57	Universal Quantized Factor
κ	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
β_i	0.603	Buoyancy coupling coefficient
H_{SCm}	0.99	SCm manifold completeness

Symbol	Value	Description
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_1_CoAnQi.cpp`, and Wolfram kernels (`uqff_kozima_kernel.wl`, `uqff_s26_kernel.wl`, `uqff_mock_theta_pi_kernel.wl`).